The paleomagnetism of the Helikian Mistastin pluton, Labrador, Canada

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A paleomagnetic study was carried out on 70 cores from 11 sites on the Helikian (1346 ± 30 m.y.) Mistastin pluton, which lies in the Churchill Province about 120 km north of the Grenville Front. Primary magnetization could not be recovered from four of the sites, which were very weakly magnetized. The magnetization of three of these was dominated by a steeply-dipping down direction, which is thought to reflect the recent earth's field direction and may have originated during the process of recent weathering. The primary magnetization was isolated from the remaining seven sites, with a mean direction of $D = 93.5°$, $I = 7.0°$ and $\alpha_{95} = 8.8°$. One site was reversely magnetized relative to the rest. The Pacific pole lies at 158.5° W and 1.0° S, with $A_{95} = 7.6°$. This pole position supports the concept of a clockwise rotation (30° ± 5°) of the northeastern Grenville Orogen during the Helikian.

Une étude paléomagnétique a été effectuée sur 70 carottes de 11 sondages dans le pluton hélikien de Mistastin (1346 ± 30 Ma), qui se trouve dans la Province Churchill, à environ 120 km du Front du Grenville. Le magnétisme primaire n'a pu être évolué dans quatre sondages qui ne montrent qu'une aimantation très faible. L'aimantation dans trois de ces sondages indique une direction à fort plongement qui est probablement celle de la direction actuelle du champ terrestre et dont l'origine serait liée à l'altération récente. Le magnétisme primaire a pu être identifié dans les sept puits restant et indique une direction moyenne pour $D = 93.5°$, $I = 7.0°$ et $\alpha_{95} = 8.8°$. Un seul site montre une direction inversée par rapport aux autres. Le pôle pacifique est à 158.5° O et 1.0° S avec $A_{95} = 7.6°$. Cette position du pôle est en accord avec le concept d'une rotation dans le sens des aiguilles d'une montre (30° ± 5°) de la partie nord-est de l'Orogène grenvillien, durant l'Hélikien.

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Introduction

Most of the Grenville Province and a north striking belt that includes the easternmost Churchill and Nain Provinces are characterized by the presence of anorthositic and quartz-monzonitic rocks (Fig. 1). The latter generally contain pyroxene, and plutons of pyroxene quartz-monzonite are particularly abundant within the anorthosite belt north of the Grenville Front. Whether or not the anorthosites are all of about the same age is still not known and, although intrusion of quartz-monzonite into anorthosite has been reported at many localities, it is not known either whether the quartz-monzonites are all significantly younger than the anorthosites. Present data suggest that regionally there may be an overlap in the ages of the anorthositic and quartz-monzonitic suites.

Much or all of the anorthosite and quartz-monzonite of the Grenville Province may have been metamorphosed since intrusion. There is no evidence, however, of late regional metamorphism on a large scale north of the Grenville Front, although the anorthosite and quartz-monzonite intrusions may themselves have produced widespread thermal effects on surrounding country rocks.

The first paleomagnetic study of an anorthosite of the eastern Shield was that by Hargraves and Burt (1967) of the Allard Lake intrusion. This body lies within the Grenville Province, and it was concluded that the observed remanent magnetization of the Allard Lake rocks was not primary, but originated during events associated with the Grenville orogeny, which ended approximately 900 m.y. ago. The Michikamau anorthosite lies to the north of the Grenville

Front and has been studied by Murthy *et al.* (1968; 1971). The primary magnetization of this body was demonstrated to reside in magnetite needles of high coercive force lying within the plagioclase crystals. The Michikamau pole position is important, as rocks of similar age from a number of localities as far away as Wyoming have yielded almost identical poles, suggesting that much of the North American Craton has not been disrupted since the intrusion of this anorthosite body.

Evidence that the Michikamau anorthosite is between 1400 and 1500 m.y. old was summarized by Murthy *et al.* (1968). Such an age is supported by a U/PB zircon determination (Krogh and Davis 1974) on quartz-monzonite from the Harp Lake complex just east of Michikamau (Fig. 1), which indicated an age of about 1450 m.y. This provides a minimum age for the Harp Lake anorthosite.

The Mistastin quartz-monzonite pluton (Fig. 1), which lies north of the Michikamau and Harp Lake intrusions, has a Rb/Sr whole-rock age of 1346 ± 30 m.y.[1] (Marchand and Crocket 1974). It is therefore significantly younger than the Michikamau intrusion, and paleomagnetic work on this undisturbed body was carried out in order to help fill the gap between the Michikamau (~1450 m.y.) and Mackenzie (~1250 m.y.) magnetic intervals (see, for example, Fahrig and Jones 1969).

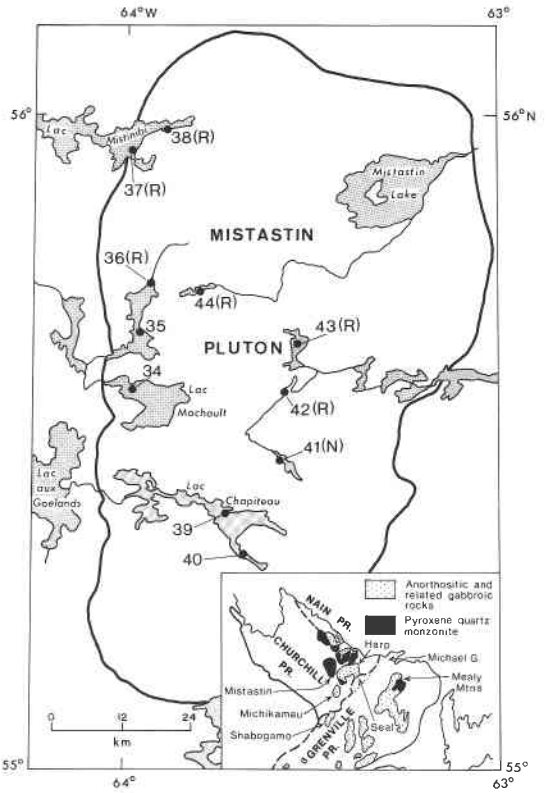


FIG. 1. Sampling site locations on the Mistastin pluton. R or N in brackets after the site number indicates that the rock of these sites is reversely or normally magnetized, respectively. Numbers lacking either designation indicate that data from these sites could not be used in the group statistics.

Geological Setting and Sampling

The Mistastin pluton has been briefly described by Taylor (1972) in his discussion of adamellite (quartz-monzonite) in northeastern Quebec and northern Labrador. The Mistastin is chiefly a coarse grained, equigranular rock that weathers a dark rust and whose fresh surface is dark greenish. Locally, however, it is highly quartzose, weathers light pinkish grey, and exhibits rapakivi texture. Taylor (1972) has presented data indicating that a similar quartz-monzonite pluton east of the Mistastin is an extremely high-level intrusion and in one locality appears to grade upward into rhyolitic material.

Seventy cores were drilled and oriented by means of a sun-compass at the 11 sites shown in Fig. 1. Difficulty was encountered in obtaining cores at several sites (notably site 42), because of the jointed and weathered nature of the rock.

Two specimens were cut, where possible, from each core sample and paleomagnetic results from each pair of specimens were averaged; resulting mean values taken to represent the sample parameters.

Measurement

Paleomagnetic measurements were made on the specimens with either a biastatic magnetometer or, when specimen magnetization was less than 5×10^{-3} A/m (5×10^{-6} emu/cm^3), a JR-3 spinner magnetometer. Generally good within-site grouping of directions of natural remanent magnetization (NRM) was observed. Figure 2 shows that most NRM site mean directions have a northerly component and dip down very steeply, suggesting an overprint of the primary

[1]The decay constant used throughout this paper is 1.39×10^{-11} y^{-1}.

magnetization of the earth's present magnetic field. While the NRM intensities of specimens showed only a limited range within each site, site mean intensities, given in Table 1, ranged more widely (2.3 to 850 $\times 10^{-3}$ A/m). Differences in the latter can be directly correlated with lithology. Weakly magnetized rock is composed of pink or grey potash feldspar-rich material, strongly magnetized, of dark greenish-brown pyroxene quartz-monzonite, while rocks of intermediate intensity have lithologies between these two extremes.

At least two specimens from each site were demagnetized progressively in alternating magnetic fields in steps of 5 mT (50 Oe) to 25 mT, using two-axis tumbling equipment. At 25 mT the intensities of some specimens had dropped to values as low as 2 $\times 10^{-4}$ A/m. Thus only the stronger specimens were further demagnetized at 35 and 45 mT. The specimens exhibited two major types of directional behaviour in response to the magnetic cleaning. Their directions either shifted progressively to a shallow-dipping easterly direction with an end point at 20 to 25 mT (or a direction reversed to this), or they maintained a steeply dipping down direction. Demagnetization curves for some of the pilot specimens are shown in Fig. 3. All the remaining specimens were demagnetized in fields of 20 and 25 mT.

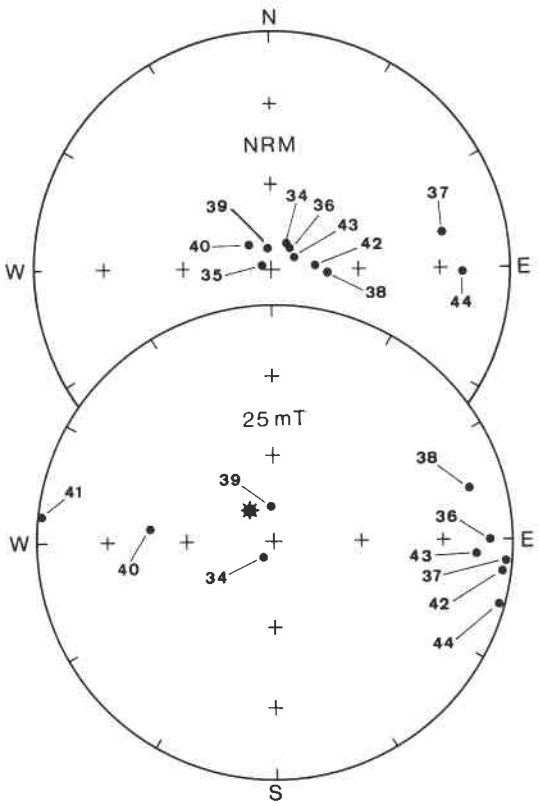


FIG. 2. Site mean directions of magnetization before (NRM) and after (25 mT) magnetic cleaning. All directions are north-seeking and downward-dipping. Star symbol indicates present earth's field direction at the Mistastin pluton.

Table 1 shows that sites 34 and 39 responded hardly at all to magnetic cleaning, their site mean directions remaining steeply down—the result, it is assumed, of recent weathering. The sample directions for site 35 were randomly distributed after treatment in both 20 and 25 mT, while four of the six samples exhibited unacceptably high θ at 25 mT; *i.e.*, the angle between the magnetization directions of the two specimens from the same sample was greater than 50°. Some of the specimen directions of site 40 moved to a shallow-dipping westerly direction at 20 and 25 mT, while the rest maintained their original steeply-dipping down direction. As only one of the samples from this site had $\theta < 50°$ (and both of the specimens from this sample had steeply-dipping down directions), it was decided to reject the data from this site as well.

It is considered that the primary (TRM) components of the remaining seven sites were successfully isolated as a result of alternating field demagnetization. The single specimen of sample 1 and the upper specimen of each of samples 2 and 7 of site 38 did not respond to magnetic cleaning, but maintained their original steeply dipping down direction. The directions of the lower specimens from samples 2 and 7 changed on demagnetization to shallow-dipping easterly directions. The directions of the specimens nearest the surface probably reflect the effects of recent weathering, while the lower specimens indicate the primary directions of magnetization of their respective samples. Thus, the data from sample 1 and from the upper specimens of samples 2 and 7 have been excluded from the site analysis. Table 1 shows that data from a small number of samples were excluded in the determination of cleaned site mean directions of four of the remaining six sites. The reason in all cases was the unacceptably high value of θ.

Grouping of sample directions about site mean

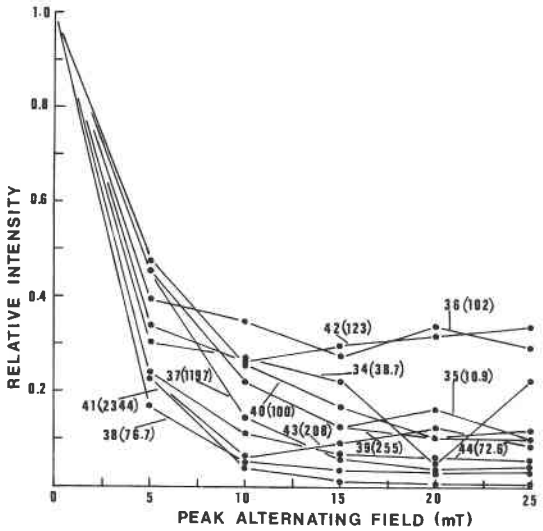


FIG. 3. Normalized alternating field demagnetization curves of one specimen from each Mistastin site. The numbers in brackets are NRM intensities in units of 10^{-3} A/m for the specimens.

directions was generally better at 25 mT than 20 mT. Thus the results for the former treatment are given in Table 1. Figure 2 shows that the site mean directions of six of the acceptable sites group well away from the direction of the present earth's field, which is typified by the mean direction of site 39. The virtual geomagnetic poles (VGPs) calculated from these site mean directions indicate that these six sites are reversely magnetized. The seventh site, 41, however, is normally magnetized; several specimens of site 40 show a similar normal magnetization. The presence of a reversal amongst the acceptable Mistastin sites supports the contention that the observed magnetization is primary. The mean direction of magnetization of the seven sites (reversing the direction of site 41 for the purpose of this calculation) is $D = 93.5°$, $I = 7.0°$, with $R = 6.876$, $k = 48.5$, and $\alpha_{95} = 8.8°$. The corresponding Pacific pole position lies at 158.6° W and 0.9° S. The mean of the VGPs falls at 158.5° W and 1.0° S, with $R = 6.906$, $K = 63.6$, and $A_{95} = 7.6°$. The Mistastin pluton is extensive and has been sampled over a large area; the pole position quoted is believed to be that of the paleomagnetic pole at the time of cooling of the pluton.

Discussion

The Mistastin pole position lies approximately

TABLE 1. Site mean directions of magnetization and pole positions for the Mistastin pluton

Site	NRM						Demagnetization in 25 mT						Site Pole		
	N	n	D	I	R	k	M	D	I	R	k	$\alpha_{95}°$	M	Lat.	Long.	Polarity
Sites from which TRM component isolated																
36	8	6	37.5	80.1	5.880	42	72	89.6	9.8	5.821	28	12.9	12	4.3° S	156.4° W	R
37	7	7	77.4	27.0	5.182	3.3	50†	95.2	2.3	6.545	13	17.3	36	2.0° N	159.0° W	R
38	7	6	92.8	71.0	5.366	7.9	18	74.5	15.7	4.977	4.9	33.8	3.4	15.3° S	145.5° W	R
41	7	6	318.6	−48.9	3.324*		670	276.5	0.1	5.781	23	14.3	29	3.7° N	158.8° W	N
42	2	2	83.0	75.3	1.978		130	98.0	3.6	1.984			33	3.0° N	161.0° W	R
43	7	6	58.6	81.6	5.829	29	92	93.5	15.6	5.913	58	8.9	24	4.6° S	160.8° W	R
44	4	2	90.7	20.7	1.745		8.4	106.1	1.9	1.960			2.9	8.2° N	167.6° W	R
Sites from which primary component not recovered																
34	7	7	28.6	79.6	6.934	92	16	209.7	83.6	6.784	28	11.6	3.2			
35	6	6	285.2	87.0	4.888	4.5	2.3	74.0	17.0	2.935*			1.1			
39	8	8	348.9	82.4	7.941	120	38	355.0	77.8	7.368	11	17.4	5.4			
40	7	7	316.0	78.9	6.968	185	51	276.2	46.6	6.428	10	19.5	4.4			

NOTES: N = number of samples (cores) collected; n = number of samples used in the analyses (see text); D = declination; I = inclination; R = resultant of n unit vectors; k = Fisher's (1953) estimate of precision; $\alpha_{95}°$ = semi-angle of cone of 95% confidence for mean direction; M = site mean intensity (in units of 10^{-3} A/m).
*These numbers refer to the 'resultants' of data that are random on Watson's (1956) criterion.
†One sample has been excluded because its intensity exceeded the mean by nearly two orders of magnitude.

13° west of the Michikamau pole, and its position relative to other North American pole positions in the age range 1450 to 1350 m.y. is shown in Fig. 4.

Pole positions such as those for rocks of the Seal Group, Michael Gabbro, or Mealy Mountains anorthosite (northwest component), do not lie on or near the apparent polar wander path indicated by poles for rocks of comparable age in the Churchill Orogen to the north, namely that for the Michikamau, Shabogamo, or Mistastin intrusions. There are a number of possible explanations for this lack of concordance. McWilliams (1974), working on rocks far to the southwest in the Grenville Province, suggested the possibility that the descending arm of the Logan Loop executes an additional loop before reaching the Franklin pole position. The

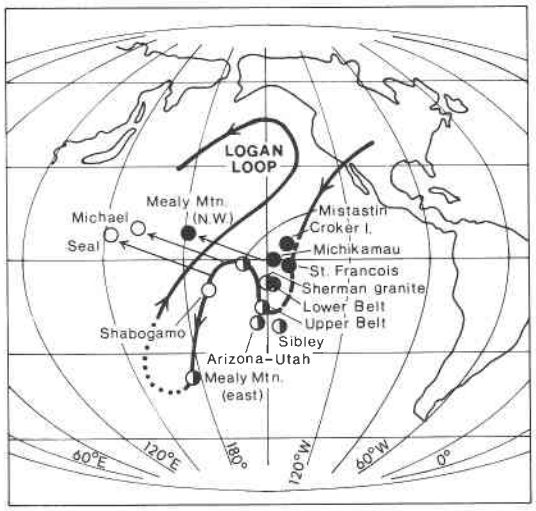


FIG. 4. North American pole positions in the age range from about 1450 to 1350 m.y. Solid symbols, units with only normal polarity; open symbols, only reversed polarity; and half shaded symbols, both normal and reversed.

References: Croker Island—Palmer 1969; Michikamau—Murthy *et al.* 1968; St. Francois—Hays and Scharon 1966; Lower and Upper Belt—Evans *et al.* 1975; Sherman Granite—Eggler and Larson 1968; Sibley—Robertson 1973; Arizona and Utah—Collinson and Runcorn 1960; Mistastin—Fahrig and Jones, this paper; Shabogamo—Fahrig *et al.* 1974; Mealy Mountain (East and N.W.)—Fahrig *et al.* 1974; Michael—Fahrig and Larochelle 1972; and Seal—Roy and Fahrig 1973.

The pole positions of the Mealy Mountain (NW), Michael, and Seal are shown with an arrow indicating the amount (30° ± 5°) and direction of rotation of these poles relative to poles of this age from outside the Grenville Orogen.

age of this subsidiary loop would presumably be between 1000 and 800 m.y. This explains the Seal, Michael, and Mealy Mountain (NW) pole positions, if all these rocks were completely remagnetized during a thermal event within the time limits 1000 to 800 m.y. ago. This does not, however, seem very likely. The Seal igneous rocks are of very low metamorphic grade, have yielded an Rb/Sr whole-rock age of 1352 ± 92 m.y. (R. K. Wanless, personal communication 1975), and show no systematic difference in magnetization across the belt. Whole-rock K–Ar ages on these rocks are close to 1000 m.y., but magnetization is almost invariably older than the whole-rock K–Ar age of basic igneous rocks (Fahrig and Jones 1969; Christie *et al.* 1975). The Michael Gabbro occurs in a zone 250 km long and has yielded an Rb/Sr age of 1487 ± 78 m.y. (R. K. Wanless, personal communication 1975). It is conceivable, but unlikely, that these rocks were remagnetized throughout their entire length 500 m.y. later, without disturbing their Sr ratios. Finally, the Mealy Mountains anorthosite is cut by unaltered diabase dikes (Gittins 1972) that have whole-rock K–Ar ages of 1100 m.y., while the gneissic country rocks of the Mealy Mountain pluton generally yield mineral K–Ar ages of about 1100 m.y. (Wanless 1970). The weight of evidence in this part of the Grenville Province therefore seems opposed to remagnetization later than 1000 m.y. ago, although it does not argue against the additional polar wandering loop suggested by McWilliams (1974), which seems to explain the pole positions of metamorphosed rocks from elsewhere in the Grenville. A second possibility, the movement of the Grenville Province back and forth through several thousand kilometres, seems to be precluded by the excellent geological fit along the Grenville Front, particularly in the region of the Seal Group. A third possibility, the rotation of this part of the Grenville about a local axis (Fahrig and Larochelle 1972; Roy and Fahrig 1973) is illustrated in Fig. 4 and is considered the best explanation.

It was originally assumed (Fahrig *et al.* 1974), because of the position of the Sibley pole (Robertson 1973), that the Shabogamo Gabbro from north of the Grenville Front might also have undergone rotation. Considerations based on pole positions and polarities of the Mistastin and of other units from the Precambrian of

North America suggest an alternative explanation. It is suggested that the apparent polar wander path from 1450 to 1350 m.y. ago trends south during a period of normal polarity (Michikamau interval), continues south, then trends north during a period of mixed polarity (the Belt interval), and then south again during a period of reversed polarity (the Shabogamo interval). This model has at least two advantages. Firstly, the Mealy Mountain (NW), Michael, and Seal poles require much less rotation ($30° \pm 5°$) than previously suggested (55°, Roy and Fahrig 1973) to bring them into agreement with poles from rocks of similar age lying outside the Grenville Orogen. Secondly, equal rotation ($30° \pm 5°$) of these poles results in agreement between their polarity and the polarity of poles from outside the Grenville.

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